CS6886 Assignment 2 - MobileNet-v2 Compression

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Question 1 - Training Baseline

1a) CIFAR-10 was loaded with the following transforms. Train: RandomCrop(32, padding=4), RandomHorizontalFlip, ToTensor, Normalize(mean=[0.4914, 0.4822, 0.4465], std=[0.2023, 0.1994, 0.2010]). Test: ToTensor and the same Normalize. No AutoAugment or Cutout was used.

1b) Model and training configuration:

Parameter	Value
Width multiplier	1.0
First conv stride	1 (modified from 2 for 32x32 CIFAR-10 input)
Classifier	nn.Linear(1280, 10) replacing default 1000-class head
Dropout	0.2 (preserved from pretrained Sequential)
Pretrained weights	ImageNet IMAGENET1K_V1
Optimizer	SGD momentum=0.9 nesterov=True wd=4e-5
Learning rate	0.05 with 5-epoch linear warmup
Scheduler	CosineAnnealingLR T_max=100
Loss	CrossEntropyLoss label_smoothing=0.1
Epochs	100
Batch size	128
Seed	42

1c) Final test top-1 accuracy: 93.59%. Loss and accuracy curves are shown in the wandb project cs6886-a2. Training loss decreased steadily from 1.03 at epoch 1 to 0.50 at epoch 100. Train accuracy reached 99.91% while test accuracy plateaued at 93.59%, indicating mild overfitting in later epochs. The primary failure mode was a 6.3% generalization gap attributable to CIFAR-10s small image size (32x32) relative to the ImageNet-pretrained feature distribution. Occasional single-epoch test accuracy dips (e.g. epoch 27: 81.41%) reflect SGD noise with high momentum at this learning rate.

Question 2 - Model Compression Implementation

2a) Post-training quantization (PTQ) with GPTQ-style layer-wise weight optimization was implemented from scratch in quantize.py without any quantization libraries.

Weight quantization: Per-channel symmetric uniform quantization for all Conv2d layers. Each output channel gets an independent scale computed as max(|W_i|) / (2^(bits-1) - 1) with zero-point fixed at zero. Per-tensor symmetric quantization for the Linear classifier.

Activation quantization: Per-tensor asymmetric uniform quantization inserted after every ReLU6 via ActFakeQuant modules. Scale and zero-point computed from percentile-clipped (99.9th percentile) running min/max over 4 calibration batches to reduce outlier sensitivity.

GPTQ: For each pointwise Conv2d (groups=1) and the Linear layer, optimal quantized weights are found by minimizing the layer-wise output error ||WX - W_q X|| using the inverse Hessian of the input activations. Error is propagated sequentially across weight elements within each output channel. Depthwise convs (groups>1) use standard PTQ.

2b) Layer quantization summary:

Layer	**Quantized**	**Weight Bits**	**Notes**
First Conv2d (features[0][0])	Yes	8	Pinned at 8-bit
Pointwise Conv2d (groups=1)	Yes	Configurable	GPTQ applied
Depthwise Conv2d (groups>1)	Yes	Configurable	Standard PTQ per-channel
BatchNorm2d	No	32 (FP32)	Kept in FP32
ReLU6 outputs	Yes	Configurable	ActFakeQuant inserted after
Classifier Linear	Yes	8	Pinned at 8-bit

2c) Storage overheads are accounted for as follows: conv weights at N*bits/8 bytes, biases at N*4 bytes (FP32), per-channel scales at out_channels*4 bytes per layer (FP32), per-channel zero-points at out_channels*4 bytes per layer (FP32), BN parameters at N*4 bytes (FP32), ActFakeQuant scale and zero-point at 2*4 bytes per activation site. At 4-bit, per-channel scales add approximately 0.13 MB overhead across all layers.

Question 3 - Compression Results

A sweep over weight_bits in {8,6,4,3,2} and act_bits in {8,6,4} was run with GPTQ enabled, yielding 15 configurations logged to wandb project cs6886-a2. The parallel coordinates chart from wandb is included below.

[SEE ATTACHED WANDB PARALLEL COORDINATES SCREENSHOT]

weight_bits	**act_bits**	**quant_acc (%)**	**weight_ratio**	**total_mb**
8	8	93.31	3.82	2.36

8	6	93.29	3.82	2.36
8	4	90.66	3.82	2.36
6	8	93.07	4.99	1.84
6	6	93.22	4.99	1.84
6	4	90.30	4.99	1.84
4	8	87.15	7.19	1.32
4	6	87.21	7.19	1.32
4	4	82.88	7.19	1.32
3	8	52.35	9.21	1.06
3	6	52.51	9.21	1.06
3	4	51.54	9.21	1.06
2	8	10.00	12.83	0.80
2	6	10.01	12.83	0.80
2	4	9.99	12.83	0.80

Question 4 - Compression Analysis

Selected configuration: W4A8 (4-bit weights, 8-bit activations) with GPTQ.

4a) Weight compression ratio: 7.19x. FP32 model weights occupy 8.53 MB. At 4-bit, quantized weights occupy 1.19 MB. Per-channel scales and zero-points add 0.13 MB overhead, giving a total compressed size of 1.32 MB.

4b) Activation compression ratio: 4.00x (theoretical: 32/8). Activations were measured as the sum of all output tensor sizes over one forward pass with batch=1, comparing FP32 (4 bytes per element) against 8-bit quantized (1 byte per element). The 4.00x ratio reflects the quantization of all ReLU6 outputs via ActFakeQuant modules.

4c) Quantized model accuracy: 87.15% at W4A8. This represents a 6.44% drop from the FP32 baseline of 93.59%. The drop is primarily attributable to depthwise conv layers which use standard PTQ without GPTQ error correction, as their small weight tensors (9 parameters per channel) make Hessian-based optimization less effective.

4d) Final approximated model size: 1.32 MB. This includes quantized weights (1.19 MB at 4-bit), per-channel scales and zero-points (0.13 MB FP32), biases (FP32), and BN parameters (FP32). The FP32 baseline is 8.53 MB.

Question 5 - Reproducibility

GitHub repository: https://github.com/Xzaos/CS6886-A2

Command	Description
python train.py	Train MobileNet-v2 for 100 epochs on CIFAR-10
python test.py --weight_bits 4 --act_bits 8 --use_gptq	Run W4A8 GPTQ quantization
python sweep.py	Run full 15-configuration sweep

Environment: Python 3.11, PyTorch 2.x, torchvision, wandb 0.28.1. Random seed: 42 (torch.manual_seed and torch.cuda.manual_seed set at the start of every script). CIFAR-10 downloads automatically on first run.